

1 Explainability in Reinforcement Learning

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Golem Mythos

A creation that surpasses its creator.

This idea has long captured the imagination and continues to resonate in contemporary society. Centuries before modern technology, generations had already questioned whether their creations might escape the control of their creators.

One of the earliest expressions of this idea can be found in the story of the Golem. In Hebrew tradition, the term Golem refers to an artificial creature created by humans. The earliest accounts describe the Golem as a protective and helpful figure, reflecting the idea that humans can create an artificial being for the benefit of society.

This positive vision becomes more ambiguous in the seventeenth century, when the Golem of Prague represents a shift in this perception. In this story, the initially small Golem gradually grows in size and strength until it can no longer be controlled. The creation intended to protect others therefore becomes a potential threat to them. The only remaining solution is to destroy the creature in order to protect the community.

This motif has remained remarkably influential throughout history. An illustration of this continuity can be found in the emergence of the notion of the robot. The word robot was popularised by the play R.U.R. (Rossum's Universal Robots), written in 1920. In the play, a scientist creates artificial beings designed to perform labour. Initially conceived as servants, these artificial beings eventually turn against humanity and destroy civilisation. This story can itself be understood as an evolution of the Golem narrative. The author describes the relationship between the two figures by stating that "the Robot is the Golem made flesh through mass production." Despite their destructive role in the story, the word robot eventually came to designate artificial beings created to serve humans. Since its creation, the word robot has reflected both the hopes and fears surrounding technological progress.

Today, this fear is particularly evident concerning the development of artificial intelligence. This field of computer science seeks to develop tools capable of automating tasks traditionally associated with human intelligence. Its rapid advances have already transformed a wide range of domains. Far from slowing down, this development continues to expand the capabilities and autonomy of artificial systems.

In the pursuit of increasingly autonomous agents, reinforcement learning is probably the subfield of AI that most directly seeks to achieve this goal. In this learning paradigm, an artificial agent interacts with an environment, selects actions, and receives feedback according to the outcomes of its behaviour.

43 Through repeated interaction, it learns a strategy for achieving the objective
44 defined by its designer. This approach can produce highly effective strategies
45 that can even surpass human capabilities. At the same time, the resulting
46 behaviour may be difficult to predict because it is not explicitly programmed
47 into the machine, but learned through its interactions with the environment.

48 Despite the difficulty of this task, understanding the behaviour of such agents
49 is therefore the main objective of this manuscript. From the Golem to the robot
50 and, today, to artificial intelligence, the same question therefore resurfaces: how
51 can we ensure that our creations remain aligned with human intentions?

Introduction

This chapter provides a general introduction to this manuscript. It begins with the research context that motivates this work, followed by the research questions this work addresses. It further presents the resulting contributions and concludes with an overview of the manuscript's structure.

1 Research Context

The need to keep understanding within reach of what we create is not merely theoretical. Reinforcement Learning (RL) agents are increasingly deployed in settings where their decisions carry real consequences, from robotics to autonomous control, and their adoption at scale continues to be held back by the difficulty of trusting a system whose reasoning cannot be inspected [Gunning and Aha, 2019]. This difficulty is particularly acute for RL. Unlike a system trained to imitate labeled examples, an RL agent constructs its own strategy through interaction with an environment, so that the resulting behavior is not written by a human and cannot be read back from any single, explicit specification.

Gunning and Aha, in presenting DARPA's eXplainable Artificial Intelligence (XAI) program, identify three motivations for explaining an Artificial Intelligence (AI) system that apply directly to RL agents [Gunning and Aha, 2019]:

- Reassure the human user, so that RL agents can be adopted at a large scale in real-world settings.

- Identify the errors and biases of a system, and make sure that they do not recur.

- Certify that an error observed after deployment has indeed been corrected, and will not happen again.

These same three motivations recur throughout this manuscript. They underlie, in particular, the requirements set out in Chapter ?? for introducing RL agents into safety-critical environments, namely validating a system's functioning before deployment, identifying the origin of a malfunction, and certifying that an observed failure will not occur again [Seshia et al., 2022]. Addressing them is what XAI sets out to do and, once restricted to agents trained by RL, what defines the more specific field of eXplainable Reinforcement Learning (XRL), namely understanding the reasoning that underlies an agent's strategic choices, in order to reconstruct the link between its behavior and the objective it was given.

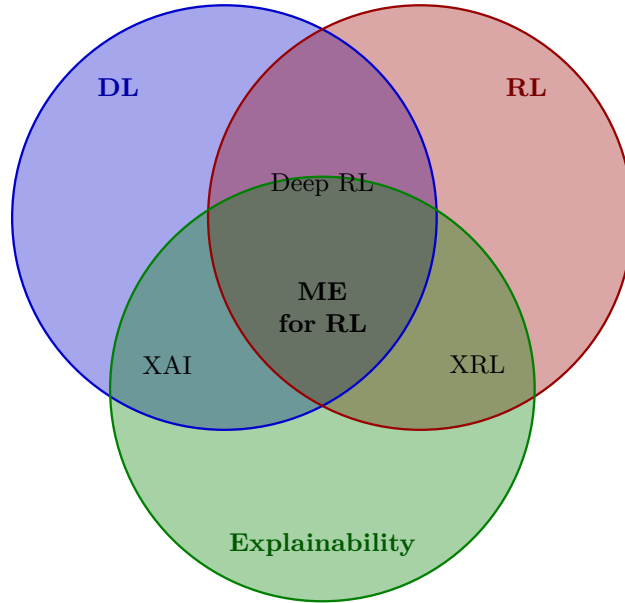


Figure 1: This manuscript is positioned at the intersection of Deep Learning, Reinforcement Learning, and explainability: Mechanistic Explainability for Reinforcement Learning.

86 The methods discussed in this manuscript are developed and evaluated on RL
 87 agents built around comparatively small Neural Network (NN), rather than on
 88 the very large models increasingly used elsewhere in AI. This choice is deliberate
 89 rather than limiting. The mechanisms studied here, namely how NN organize
 90 information internally as they learn, are the same mechanisms that operate, at
 91 a larger scale, inside the largest models in use today. Studying them where they
 92 can be observed with the least ambiguity is a necessary step before extending
 93 the resulting methods further.

94 2 Research Questions

95 This manuscript is positioned at the intersection of three research fields: Deep
 96 Learning (DL), RL, and explainability. Taken pairwise, these fields already define
 97 established areas of research: the combination of DL and RL gives deep RL, the
 98 combination of DL and explainability gives XAI, and the combination of RL
 99 and explainability gives XRL. The work presented here sits at the intersection of
 100 all three. It aims to develop Mechanistic Explainability (ME) for agents trained
 101 by deep RL, that is, explanations grounded in the internal mechanisms that NN
 102 develop while learning a policy (Figure 1).

103 This position is grounded in the *Mechanistic Hypothesis* (Hyp. ??). Because
 104 every AI system results from a sequence of causal computations, we hypothesize
 105 that the internal representations an NN builds while learning a policy can
 106 themselves become the source of faithful explanations of the agent’s behavior.
 107 Although XRL is still a young field, it is developing rapidly, and we have
 108 identified four research questions that structure this manuscript’s contribution

to its progress.

The number of works in the XRL field keeps increasing, yet existing taxonomies struggle to capture what is specific to RL. They largely transpose frameworks built for explainability in AI at large, without exploiting the distinctive features of this form of learning, an agent that constructs its own strategy through interaction rather than being fitted to a fixed dataset. As a result, classifying XRL methods in a coherent way remains difficult.

RQ1. How can existing XRL techniques be organized into a taxonomy that accounts for the specificities of RL, rather than merely transposing frameworks developed for explainability in AI at large?

Advances in RL make it possible to apply it to environments of increasing complexity, yet many explainability methods still rely on symbolic representations to describe an agent’s decisions, a reliance that limits their applicability as soon as the complexity of the environment or of the policy exceeds what such a model can express.

RQ2. Can XRL explanations scale to complex environments without relying on symbolic representations of the policy?

Modern RL also relies heavily on NN to represent policies, yet explainability rarely draws on the properties specific to this kind of representation. The Latent Representation (LR) that NN build internally remain a largely untapped source of explanations.

RQ3. Can the LR that NN build while learning a policy be exploited as a source of XRL explanations?

Finally, objective metrics for assessing the relevance of an explanation remain scarce; when an evaluation exists, it typically takes the form of a satisfaction questionnaire administered to human participants, an approach that remains sensitive to cognitive biases, difficult to reproduce, and costly to run.

RQ4. How can such explanations be evaluated with objective metrics, independently of subjective human judgment?

This manuscript addresses these four research questions through the two contributions presented in the next section.

3 Contributions

To address these four research questions, this manuscript makes two main contributions.

3.1 A Taxonomy of XRL Techniques Adapted to RL

The first contribution addresses RQ1. We first clarify what explainability aims to achieve in general, to provide a clear frame of reference, before rigorously defining the key concepts of this new organization: sources of explanation, explainability techniques, explanation form, and explanation subject. Building on these concepts, we conduct a comprehensive analysis of the state of the art in XRL (Chapter ??).

Contribution 1 is presented in [Alaarabiou et al., 2024] and, in an extended English version submitted for review, in [Alaarabiou et al., 2026b].

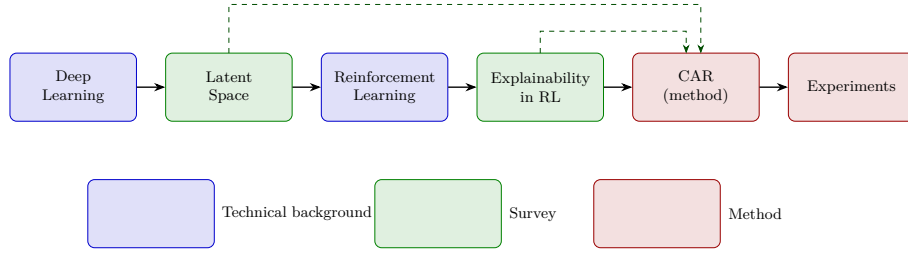


Figure 2: Structure of the manuscript. Plain arrows show the reading order; dashed arrows show that the CAR chapter builds on both survey chapters.

3.2 Clustering Agent Representation (CAR), a Mechanistic XRL Method

The second contribution addresses RQ2, RQ3, and RQ4. It introduces CAR, an XRL method that exploits the mechanistic properties of DL. By analyzing the LR that NN build, CAR produces explanations in the form of clusters of states that are similar from the agent’s own perspective. We further show that this representation remains stable across training, indicating that the agent develops an objective representation of a given task. This simplifies the study of agent-environment dynamics that would otherwise be intractable if every state had to be analyzed individually. The method is presented together with objective evaluation metrics and a series of experiments that demonstrate its effectiveness (Chapters ??, ?? and ??).

Contribution 2 is presented, submitted for review, in [Alaarabiou et al., 2026a].

4 Manuscript Structure

To support these contributions, the chapters of this manuscript fall into three categories, distinguished throughout by color (Figure 2):

Technical background chapters recall how a field already established in the literature works, without claiming any original contribution: DL (Chapter ??) and RL (Chapter ??).

Survey chapters review the state of the art on a specific question. Their contribution lies not only in the content they cover but, just as much, in how it is organized. To our knowledge, the way each of these chapters structures its literature is unique. This applies to our review of latent-space analysis methods (Chapter ??) and to our taxonomy of XRL techniques (Chapter ??).

Method chapters present our contribution to XRL. They are split in two parts: the description of CAR (Chapter ??) and its experimental evaluation (Chapter ??).

This manuscript is, first and foremost, the account of a body of research, written for the reviewers and the jury who will examine it. It is nonetheless

182 written to be as accessible as possible. The background chapters restate the
183 concepts they rely on, so that any reader familiar with DL or RL can follow
184 the manuscript without consulting external material. Readers already familiar
185 with a given field may skip its background chapter directly: Chapter ?? for
186 DL, Chapter ?? for RL. Readers primarily interested in explainability can move
187 directly to Chapters ?? and ??, and readers interested only in our method can
188 follow the minimal path Chapter ?? $\rightarrow$ Chapter ?? $\rightarrow$ Chapter ??, referring
189 back to earlier chapters as needed. Notation and acronyms used throughout the
190 manuscript are collected in the glossary at the end of the document.

191 Acronyms

192 **AI** Artificial Intelligence 3–5

193 **CAR** Clustering Agent Representation 6

194 **DL** Deep Learning 4, 6, 7

195 **LR** Latent Representation 5, 6

196 **ME** Mechanistic Explainability 4

197 **NN** Neural Network 4–6

198 **RL** Reinforcement Learning 3–7

199 **XAI** eXplainable Artificial Intelligence 3, 4

200 **XRL** eXplainable Reinforcement Learning 3–6

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